

# **CAREER EPISODE 1**

## **Fire and metal detection robot**

### **A. INTRODUCTION**

#### **CE 1.1**

While pursuing graduation in engineering I executed the project “Fire and metal detection robot” wherein I showcased my electronics engineering domain knowledge. Specifically in this project working environment I worked on the designing & development of a robot that can detect the fire and metal in mines with the deployment of sensor technology and microcontroller (AT89C51). Using the best technologies and engineering approaches during the project, I was proficiently able to conduct robot designing and interfacing of the associated components with the controller. I led teamwork being responsible as the team leader and worked sincerely to deliver work.

### **B. BACKGROUND**

#### **CE 1.2**

For the project designing part I reviewed various embedded systems and other technologies which are deployed for controlling system operations with higher efficiency and accuracy. Technically I found that the microcontroller is one of the systems that perform desired control operations, however such controllers are mainly programmed for performing the desired task. In context to my project analysis, I also studied the sensor technologies which I considered for detecting the presence of smoke/fire and metals. With this review I gained a proper understanding about the same and also I referred to the procedure involved in the interfacing of the sensor technology module with microcontroller (AT89C51). I studied thoroughly about all the technological modules that needed to be deployed for the optimized design of the proposed robot for the fire and metal mine detection. As per my project designing concept I configured microcontroller AT89C51 as primary controller and the sensor modules used for sensing the

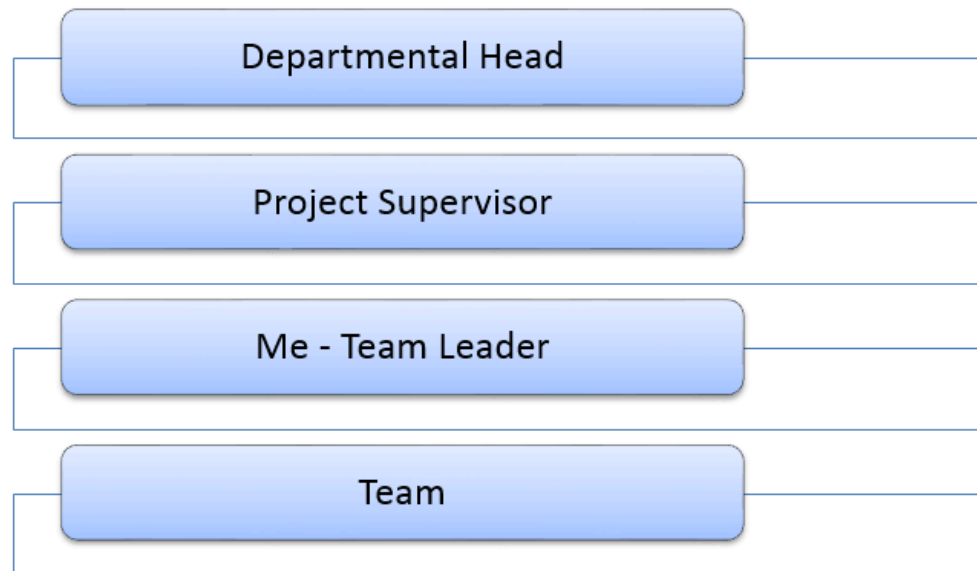
metal objects and fire. As per my project plan I worked hard for the designing & development of a robot that can detect the fire and metal in mines with the deployment of sensor technology and microcontroller (AT89C51).

### **CE 1.3**

- ◆ Specifically in this project working environment I worked on the designing & development of a robot that can detect the fire and metal mines with the deployment of sensor technology and microcontroller (AT89C51).
- ◆ In the project plan I specified the project designing and development procedure that included both software and hardware configuration in terms of circuit design, circuit simulation and hardware circuit testing analysis.
- ◆ For the purpose of fire detection I selected a flame sensor and specifically I designed a metal detection circuit using 555-timer IC & R-L-C circuit components whereas both these sensor modules were interfaced to the microcontroller AT89C51 based on its pin-configuration. Additionally I integrated the LCD for displaying significant information and also the buzzer was provided, however two motors along with a motor driver IC for driving the robot.
- ◆ For the aim of designing the circuit I worked under the circuit designing & simulation environment of Proteus software package however specific components of the circuit were selected and represented with standard symbols.
- ◆ Following the programming algorithm and process, I interfaced the system components with the microcontroller AT89C51. Most importantly I considered KEIL as an standard IDE with Embedded C as the programming language to program the microcontroller and after compiling the program I obtained the HEX file which I dumped into the microcontroller.
- ◆ With reference to the programming of AT89C51 microcontroller, I accomplished the circuit interfacing in Proteus software package based on which the simulation executed to verify the operations after providing the power.
- ◆ In the last phase I was intentionally involved in the development of a hardware circuit model of the robot and for this general breadboard used for placing the circuit

components. I provided the required power supply to the breadboard and the working operations of the robot demonstrated.

#### **CE 1.4**



### **C. PERSONAL ENGINEERING ACTIVITY**

#### **CE 1.5**

As per my project aim I utilized this project working environment for the designing & development of a robot that can detect the fire and metals in the mining area with the deployment of sensor technology and microcontroller (AT89C51). In the project plan I specified the project designing and development procedure that included both software and hardware configuration in terms of circuit design, circuit simulation and hardware circuit testing analysis. In accordance to my designing concept I depicted here in figure 1, the distinct system units which I integrated for system architecture-

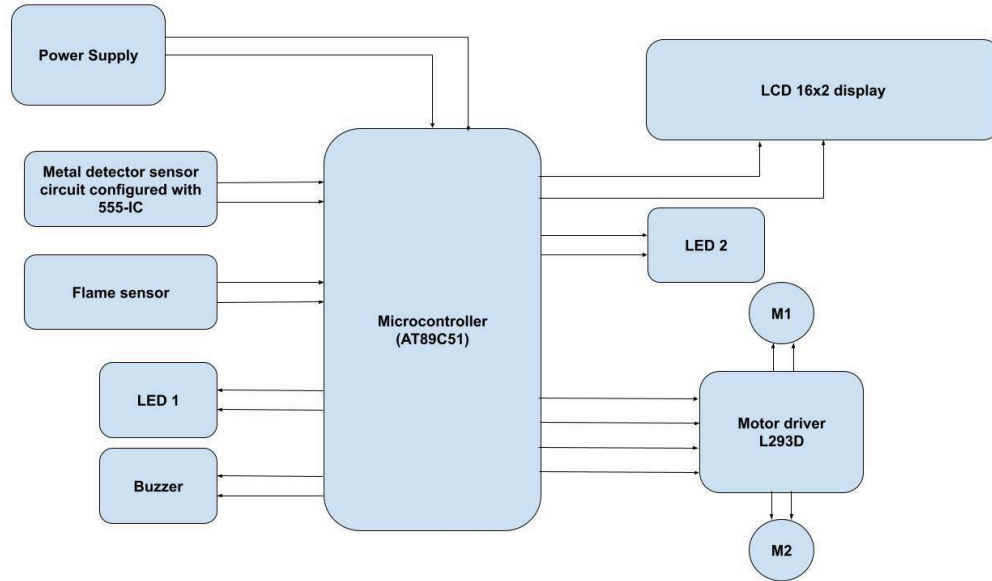


Fig. 1: System architecture

For the purpose of fire detection I selected a flame sensor, however specifically I designed a metal detection circuit using 555-timer IC and R-L-C circuit components. Further both these sensor modules were interfaced to the microcontroller AT89C51 based on its pin-configuration. Additionally I integrated the LCD for displaying significant information and also the buzzer was provided, however to drive the robot two motors along with a motor driver (IC L293D) integrated to the microcontroller. With the microcontroller, I also provided a time delay in the system program for reset and continuation of the operations. For the microcontroller (AT89C51) programming, Embedded C language was utilized under the KEIL i.e. an integrated development environment of microcontroller. In all I deployed AT89C51 microcontroller as the primary component for overall system control whereas voltage regulator IC based power supply circuit also included in the system designing analysis. For the aim of designing the circuit I worked under the circuit designing & simulation environment of Proteus software package however specific components of the circuit were selected from the libraries and represented with standard symbols.

## CE 1.6

With the use of Proteus software package I worked on circuit designing part wherein I selected a flame sensor from the tool library for the purpose of detection of fire , however for the detection of the metals I specifically designed a metal detection circuit using 555-timer IC & R-L-C circuit

components and further I interfaced both these sensor modules to the microcontroller AT89C51 based on its pin-configuration. Apart from this I also interfaced the LCD with the microcontroller for displaying system status and technically I utilized a preset which would behave like the variable voltage device. With the variation in voltage, the preset would also vary and the LCD contrast could be adjusted and further I utilized a variable resistor as the potentiometer. In addition to this using a transistor I connected a buzzer with the microcontroller to produce sound alert when fire or metal detected and for the movement of the robot two motors connected with the microcontroller using a motor driver IC. Another significant part in the project design was programming of the microcontroller and therefore I first created a operational flowchart i.e. given here below in figure 2-

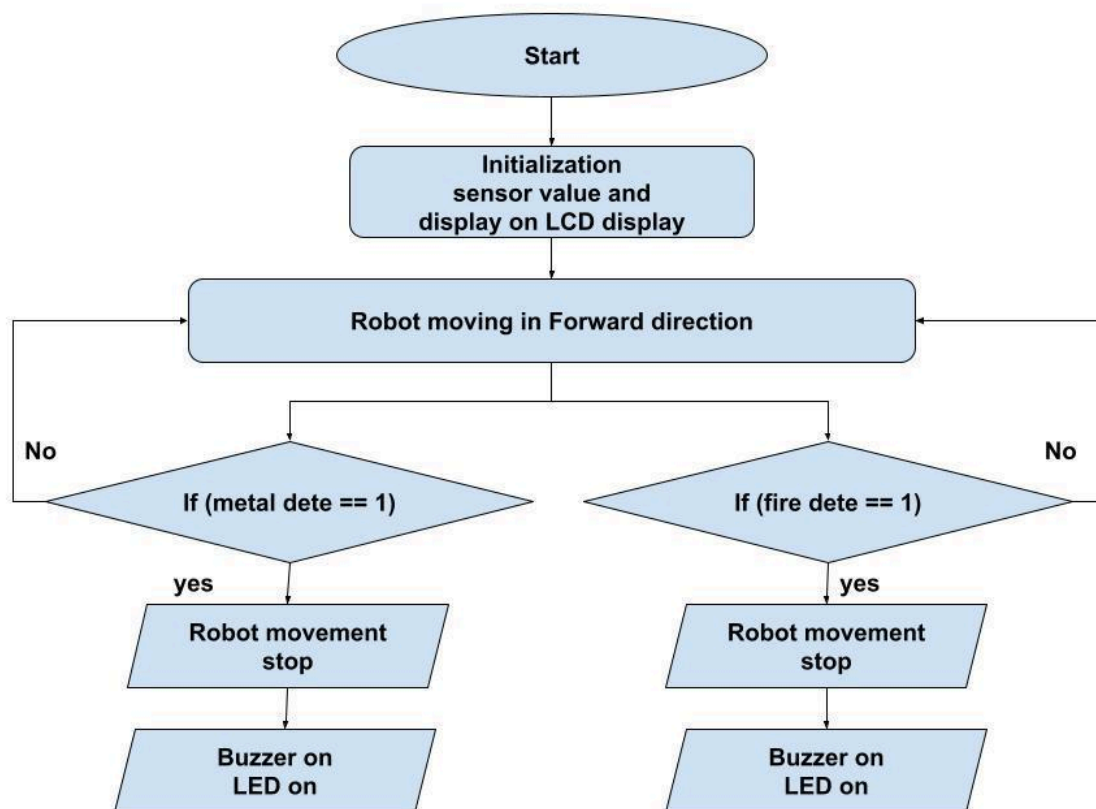


Fig. 2: Flowchart

Based on this I defined the programming algorithm and process corresponding to which I interfaced the system components with the microcontroller AT89C51 as per the standard procedure. Most importantly I considered KEIL as an standard IDE with Embedded C as the

programming language to program the microcontroller and after compiling the program I obtained the HEX file which I dumped into the microcontroller. In following images I illustrated the programming part carried out-

```
void forward();
void stop();

int rec;

char abc[16];

void L_CMD (char L_value)
{
    L_RS_PIN = 0;
    L_EN_PIN= 1;

    L_DATA_P= L_value;

    L_RS_PIN = 0;
    L_EN_PIN = 0;
}

void fun_delay (unsigned int a )
{
    char k; char l;

    for (k=0;k<a;k++)
    {
```

Fig. 3: Programming part

```

void fun_delay (unsigned int a )
{
    char k; char l;

    for (k=0;k<a;k++)
    {

        for (l=0;l<10;l++)
        {

        }

    }
}

void LCD_begin (void)
{
    L_CMD(0x30);
    fun_delay(30);

    L_CMD(0x38);
    fun_delay(30);

    L_CMD(0x0c);
    fun_delay(30);
}

```

Fig. 4: Programming part

```

void LCD_begin (void)
{
    L_CMD(0x30);
    fun_delay(30);

    L_CMD(0x38);
    fun_delay(30);

    L_CMD(0x0c);
    fun_delay(30);

    L_CMD(0x01);
    fun_delay(30);

    L_CMD(0x06);
    fun_delay(30);
}

void DIS_DATA(int L_value)
{
    L_RS_PIN=1;

    L_EN_PIN = 1;    //Strobe the enable pin

    L_DATA_P= L_value;    //Put the value on the pins

    L_RS_PIN = 1;
}

```

Fig. 5: Programming part

```
void DIS_DATA(int L_value)
{
    L_RS_PIN=1;

    L_EN_PIN = 1;    //Strobe the enable pin

    L_DATA_P= L_value;    //Put the value on the pins

    L_RS_PIN = 1;
    L_EN_PIN = 0;
}

void disp_fun (unsigned char curSor, unsigned char *point)
{
    L_CMD(0x80 | curSor);
    fun_delay(1);
    while(*point)
    {
        DIS_DATA(*point++);
    }
}

int ser_rec(void)
{
    while(!RI);
}
```

Fig. 6: Programming part



```

    L_DATA_P= L_value;      //Put the value on the pins

    L_RS_PIN = 1;
    L_EN_PIN = 0;
}

void disp_fun (unsigned char curSor, unsigned char *point)
{
    L_CMD(0x80 | curSor);
    fun_delay(1);
    while(*point)
    {
        DIS_DATA(*point++);
    }
}

int ser_rec(void)
{
    while(!RI);
    RI=0;
    return SBUF;
}

```

Fig. 7: Programming part

```

LCD_begin();

disp_fun(0x00," Fire and Metal ");
disp_fun(0xc0,"Detection System");
fun_delay(2000);

    while(1)
    {
        forward();
        if(fire_sen)
        {
            stop();
            buzz = 1;
            disp_fun(0x00,"Metal detected !");
            disp_fun(0xc0," Buzzer ON ");
            fun_delay(2000);
            disp_fun(0x00,"Robot direction ");
            disp_fun(0xc0," Stop ! ");
            fun_delay(2000);
        }

        if(metl_sen)
        {
            stop();
            buzz = 1;

```

Fig. 8: Programming part

```
if (fire_sen)
{
    stop();
    buzz = 1;
    disp_fun(0x00, "Metal detected !");
    disp_fun(0xc0, "    Buzzer ON    ");
    fun_delay(2000);
    disp_fun(0x00, "Robot direction ");
    disp_fun(0xc0, "    Stop !    ");
    fun_delay(2000);
}

if (metl_sen)
{
    stop();
    buzz = 1;
    disp_fun(0x00, " Fire detected !");
    disp_fun(0xc0, "    Buzzer ON    ");
    fun_delay(2000);
    disp_fun(0x00, "Robot direction ");
    disp_fun(0xc0, "    Stop !    ");
    fun_delay(2000);
}
```

Fig. 9: Programming part

```

        disp_fun(0x00," Fire detected !");
        disp_fun(0xc0,"  Buzzer ON  ");
        fun_delay(2000);
        disp_fun(0x00,"Robot direction ");
        disp_fun(0xc0,"      Stop !      ");
        fun_delay(2000);
    }

    else
    {
        buzz = 0;
    }
}

void forward()
{
    disp_fun(0x00," Fire and Metal ");
    disp_fun(0xc0,"Detection System");
    fun_delay(2000);
    moto1 = 1;
    moto2 = 0;
    moto3 = 1;
    moto4 = 0;
}

```

Fig. 10: Programming part

With reference to the programming of AT89C51 microcontroller, I accomplished the circuit interfacing in Proteus software package, however significant libraries were added as per the requirements. I followed the procedure for designing the circuit in Proteus and based on which the simulation executed to verify the operations after providing the power. Therefore the circuit design and simulation analysis which I conducted in Proteus illustrated here as follows-



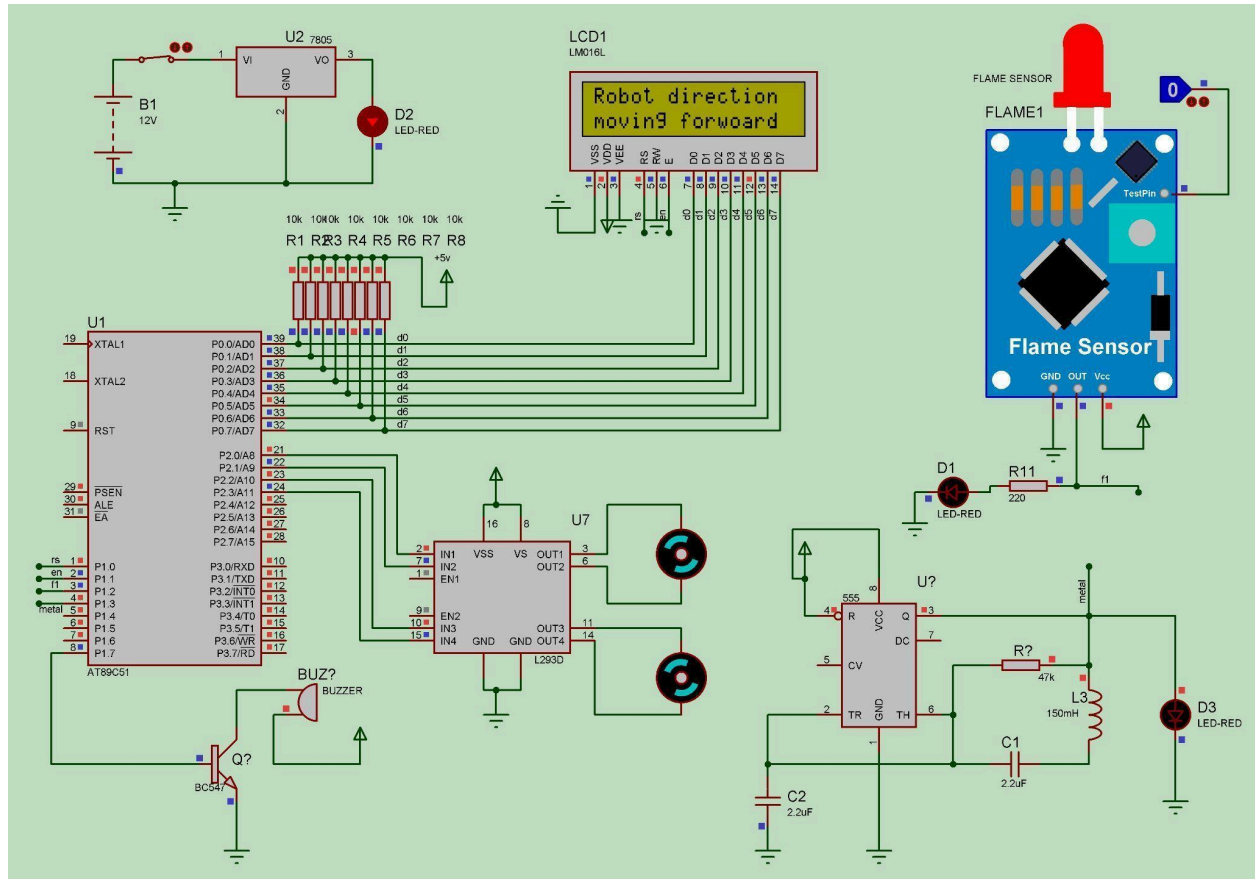


Fig. 12: Simulation part

Further flame sensor detects fire and sends input to the microcontroller however on detecting any fire the system stops the movement of robot and simultaneously alerts through buzzer as depicted in below figure 13-

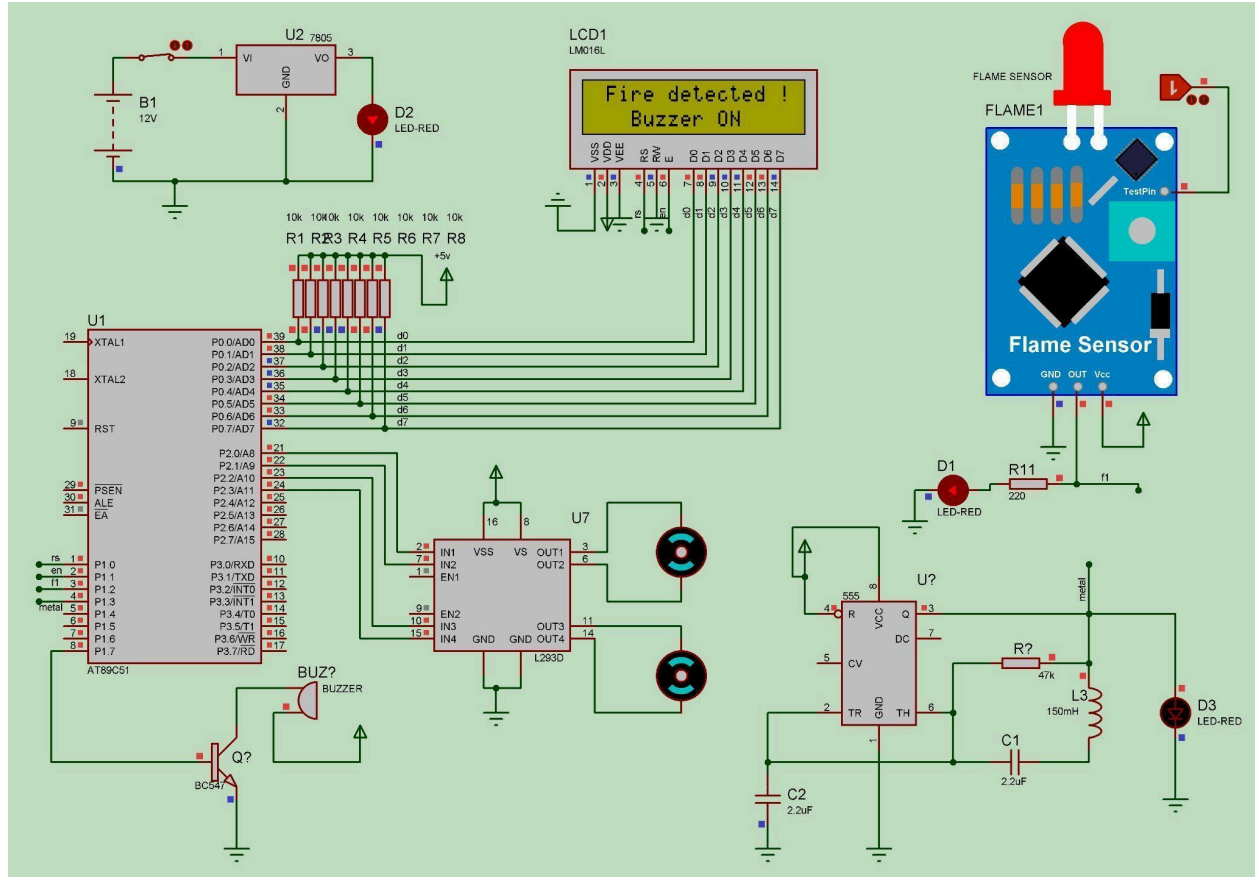


Fig. 13: Simulation part

For the metal detection part I selected a 555-timer IC as pulse signal generator whereas using the resistor ( $R=47K\Omega$ ), inductor ( $L=150mH$ ) and capacitor ( $C=2.2\mu F$ ) I created the circuit as shown in figure 13. With this circuit configuration whenever the metal piece kept near the detection circuit the value of inductance changes due to which overall impedance of the R-L-C circuit changes i.e. main working principle of the circuit. Further this variation read by the controller and it stops the robot movement and simultaneously buzzer on, LED on along with LCD displayed the metal detected which I illustrated in below simulation in figure 14-

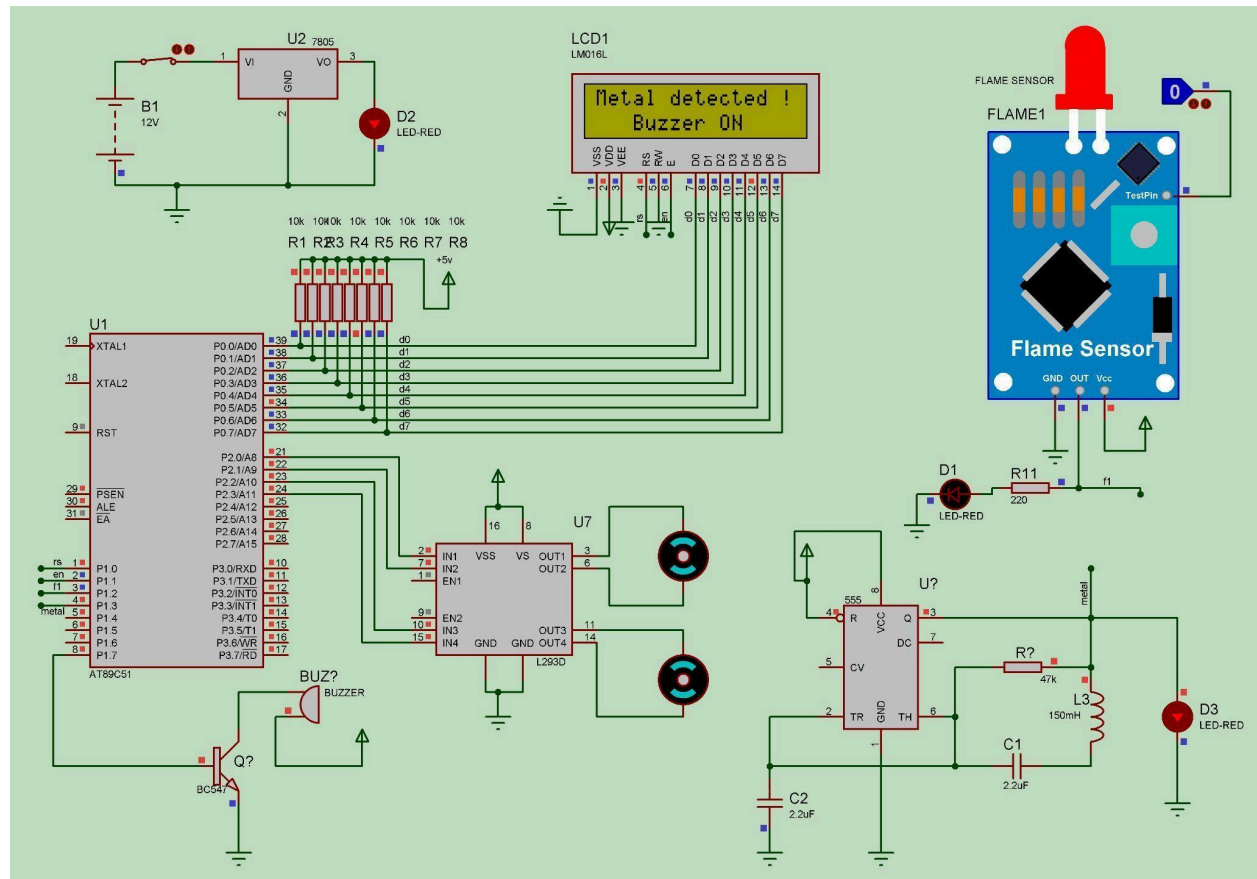


Fig. 14: Simulation part

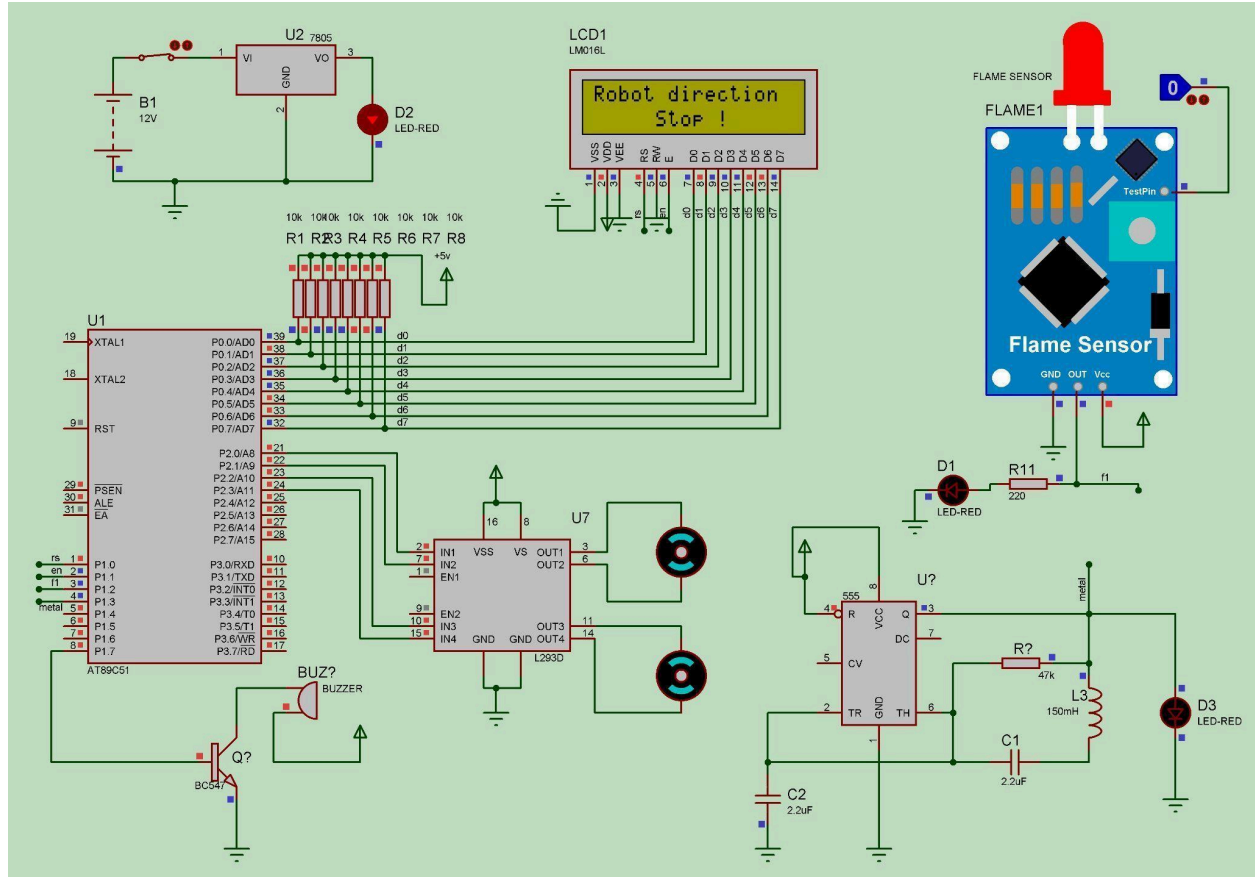


Fig. 15: Simulation part

In reference to the simulation I followed the working operation of the designed robot to validate the circuit design and therefore fire detection was carried out with the use of a flame sensor and using a 555-timer IC, a metal detection circuit configured. By following the proper process, I interfaced the system components with the controller. For example I utilized a motor driver IC L293D and through proper circuit configuration I deployed two DC motors for driving the robot in the desired direction. Finally I validated the circuit configuration with simulation analysis and for this I first initialized the system due to which the robot started moving in a forward direction whereas to confirm the detection of fire and metal I implemented these two with LEDs which glow when the detection takes place, however simultaneously the buzzer gets on and the LCD displays the respective detection. According to the designed circuit I specified that the buzzer was provided and interfaced with the controller in such a way that this buzzer would be triggered when the microcontroller would provide a signal based on the inputs received from the sensors deployed namely fire and the metal sensor.



For the hardware interfacing part I placed and interfaced all the electronic components used systematically in the circuit and achieved desired robot design as well as the framework. So, in the last phase I was intentionally involved in the development of a hardware circuit model of the robot and for this general breadboard used for placing the circuit components. Therefore, I provided the required power supply to the breadboard and the working operations of the robot demonstrated. The hardware circuit modeling and working demonstration shown here as follows-

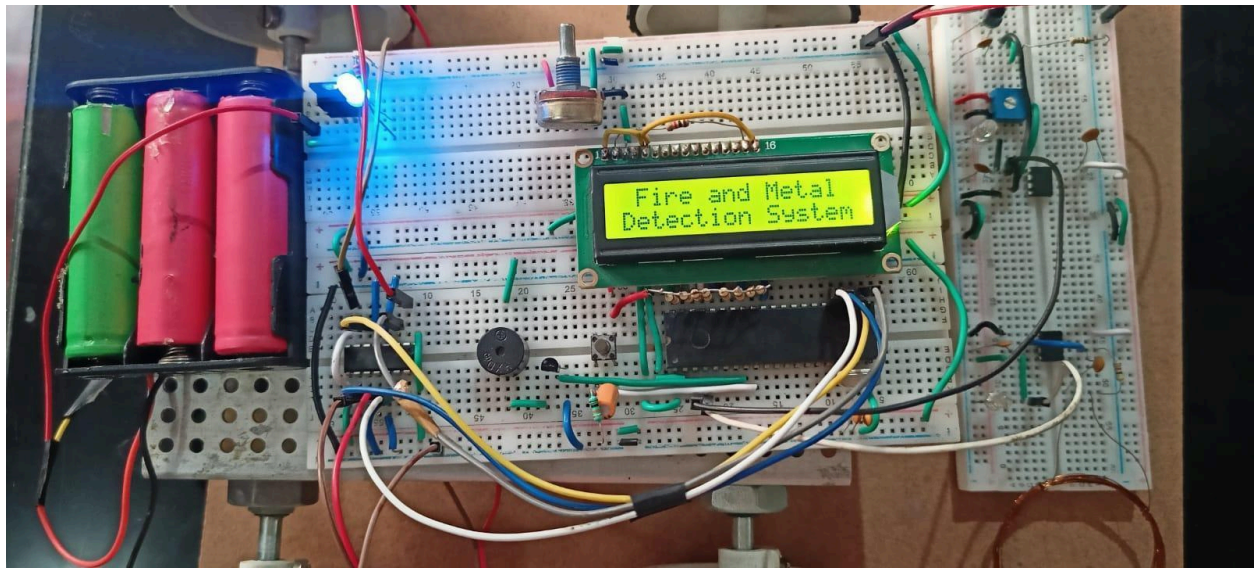


Fig. 16: Hardware interfacing part

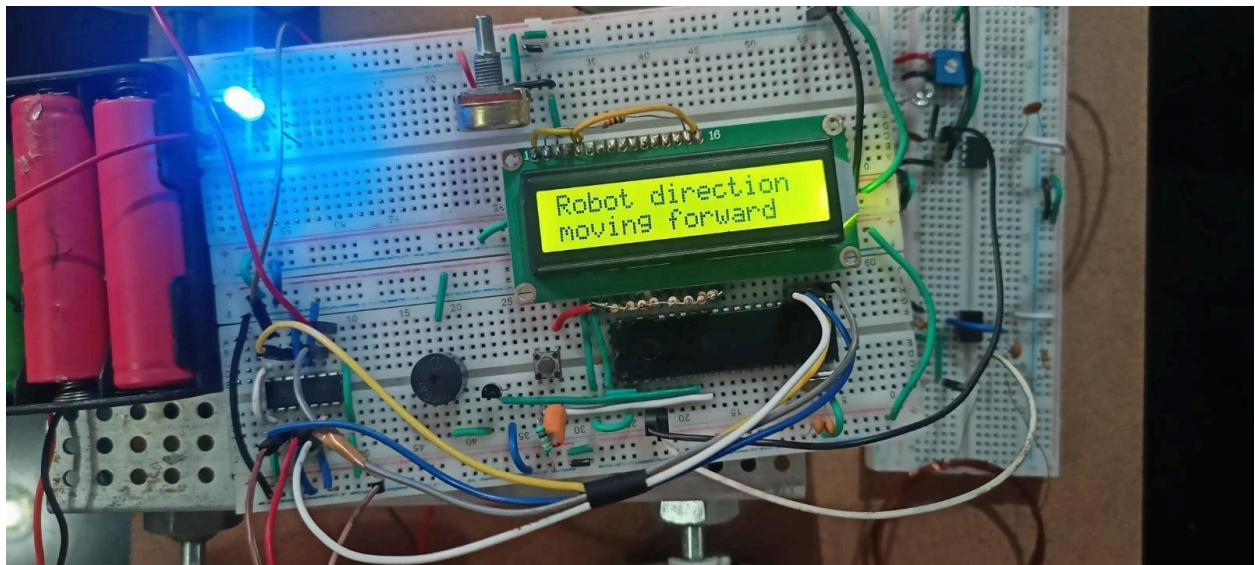


Fig. 17: Hardware interfacing part

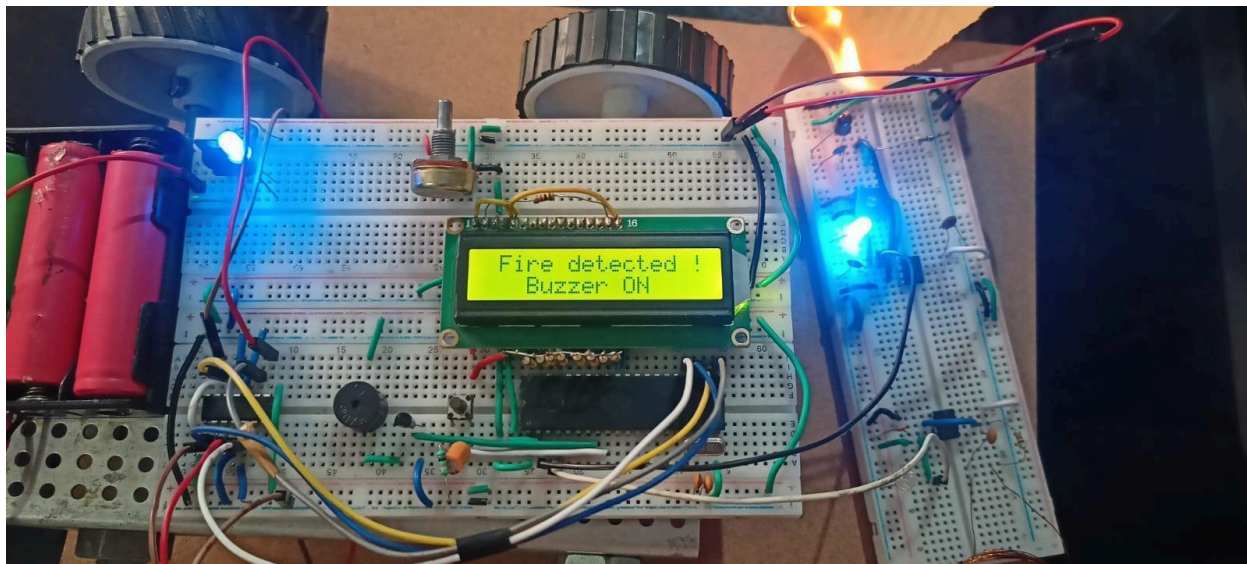


Fig. 18: Hardware interfacing part

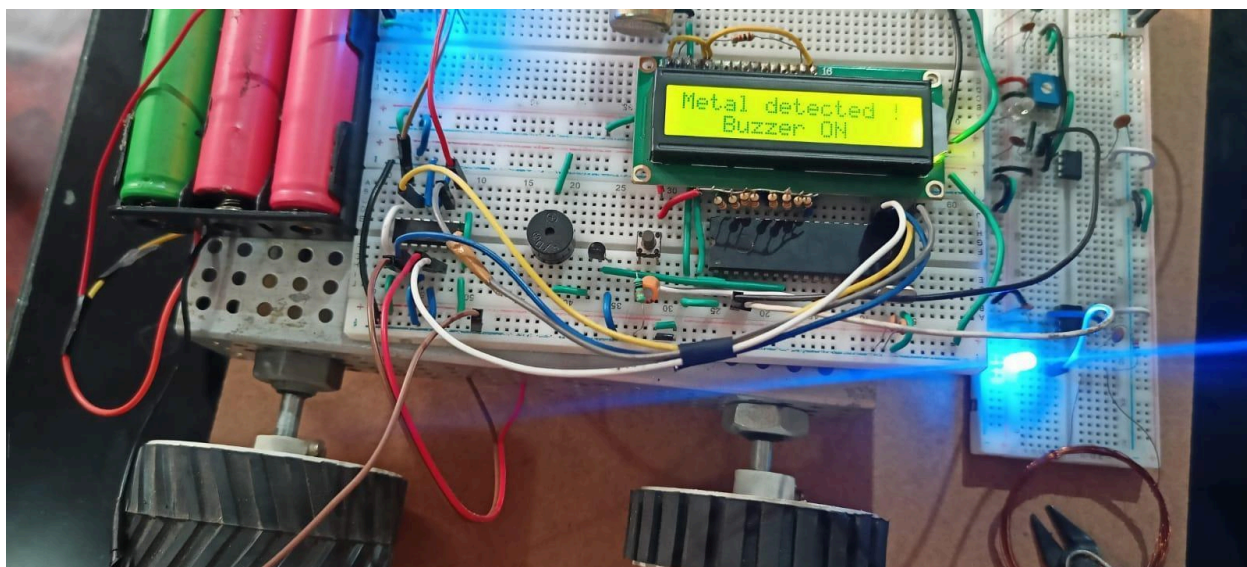


Fig. 19: Hardware interfacing part



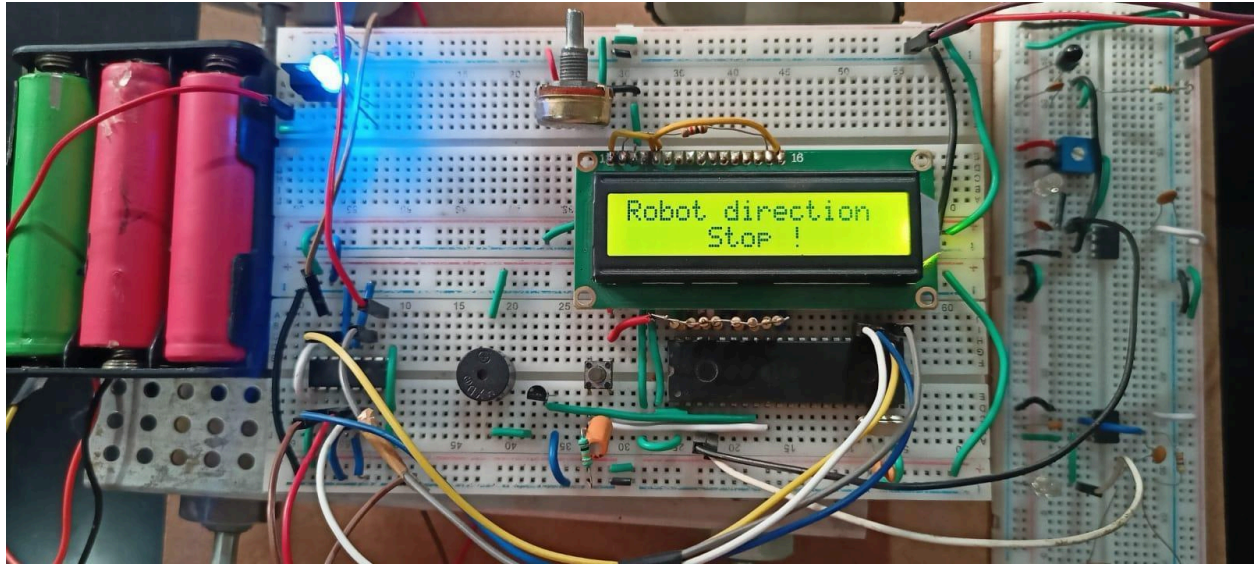


Fig. 20: Hardware interfacing part

In this way I developed a hardware circuit model and based on the supplied power I evaluated the operations with respect to the simulation. Finally I completed the design and development of a prototype model for the proposed fire and metal detection robot. With the designed system, manpower would be reduced and accuracy, as well as mine detection efficiency, could be improved.

### CE 1.8

I overcame the unnecessary issues encountered during the system design with my technical knowledge and electronics engineering skills. For instance designing a metal detection circuit using 555-timer was not easy for me and therefore initially I thoroughly studied the pin-specifications of the 555-timer IC based on which required circuit connections were performed using an R-L-C for the detection of metal. I specified the values of the R-L-C components and accordingly circuit interfacing performed. After this I integrated the designed metal detection circuit with AT89C51 microcontroller at port 1. In this way I handled the robot design task with efficiency and overcame associated technicalities with my knowledge and problem-solving ability. For example the configuration of circuit elements and their connections created some problems during the circuit designing process in Proteus. So to overcome this issue I followed the standard datasheets of circuit elements and accordingly all connections performed

by me. Further my programming-related knowledge helped me in proper controller coding and accomplishing optimal framework with complexities. By working step by step, I was able to reduce the shortcomings and hindrances manifolds throughout the span of work. My teammates used their potential fully and handled difficulties to great extent.

## **D. SUMMARY**

### **CE 1.9**

My project head helped me in achieving the defined targets by supervising my work and by advising me at each stage whenever needed. I considered the undertaken project work as a tremendous chance to improve my engineering knowledge and skills. By investigating the articles and literature contents, project-related information was compiled which aided me in understanding the work. I learned about various strategies and procedures during the execution of tasks which also increased my contextual knowledge. I opted for innovative techniques and deployed them for conducting the tasks of the project. I kept in coordination during the meetings and helped my team members in sorting out the project issues encountered while carrying out the work. I adhered to the protocols and norms underpinning the engineering institution which resulted in the fulfillment of legal requirements. My team members and I worked together to carry out the project accurately and to meet the given targets perfectly. I professionally accomplished the tasks and activities and delivered the work within the allocated deadline.

### **CE 1.10**

I designed a robot for fire and metal detection using an AT89C51 microcontroller during this current work that was the collective efforts of the team and me. I worked meticulously applying my electronics engineering skills fully for the efficient completion of the project, robot designing. My software skills also proved helpful in working in an integrated development environment for the optimal controller coding. By performing the robot design-related tasks step by step, I was able to accomplish the work and all the project phases in the desired manner. I modeled a supply unit for the operations of the controller as well as other components deployed in the system design. I interfaced the components used with the microcontroller so that the expected framework could be achieved. The system could be efficiently deployed for detecting

fire and metal mines. Due to automated robotic operations, accuracy and efficiency in detecting fire and mine would be high. My team and I made the best possible attempts for the project design. My team contributed significantly in the completion of the work optimally.